Kernel Regression

Given a training dataset $\{x_i, y_i\}_{i=1}^n$, kernel regression approximates the unknown nonlinear relation between x and y with a function of form

$$y \approx f(x; w) = \sum_{i=1}^{n} w_i k(x, x_i),$$

where $k(x, x')$ is a positive definite kernel specified by the users, and w_i is a set of weights. We will use the simple Gaussian radial basis function (RBF) kernel,

$$k(x, x') = exp(-\frac{||x - x'||^2}{2h^2}),$$

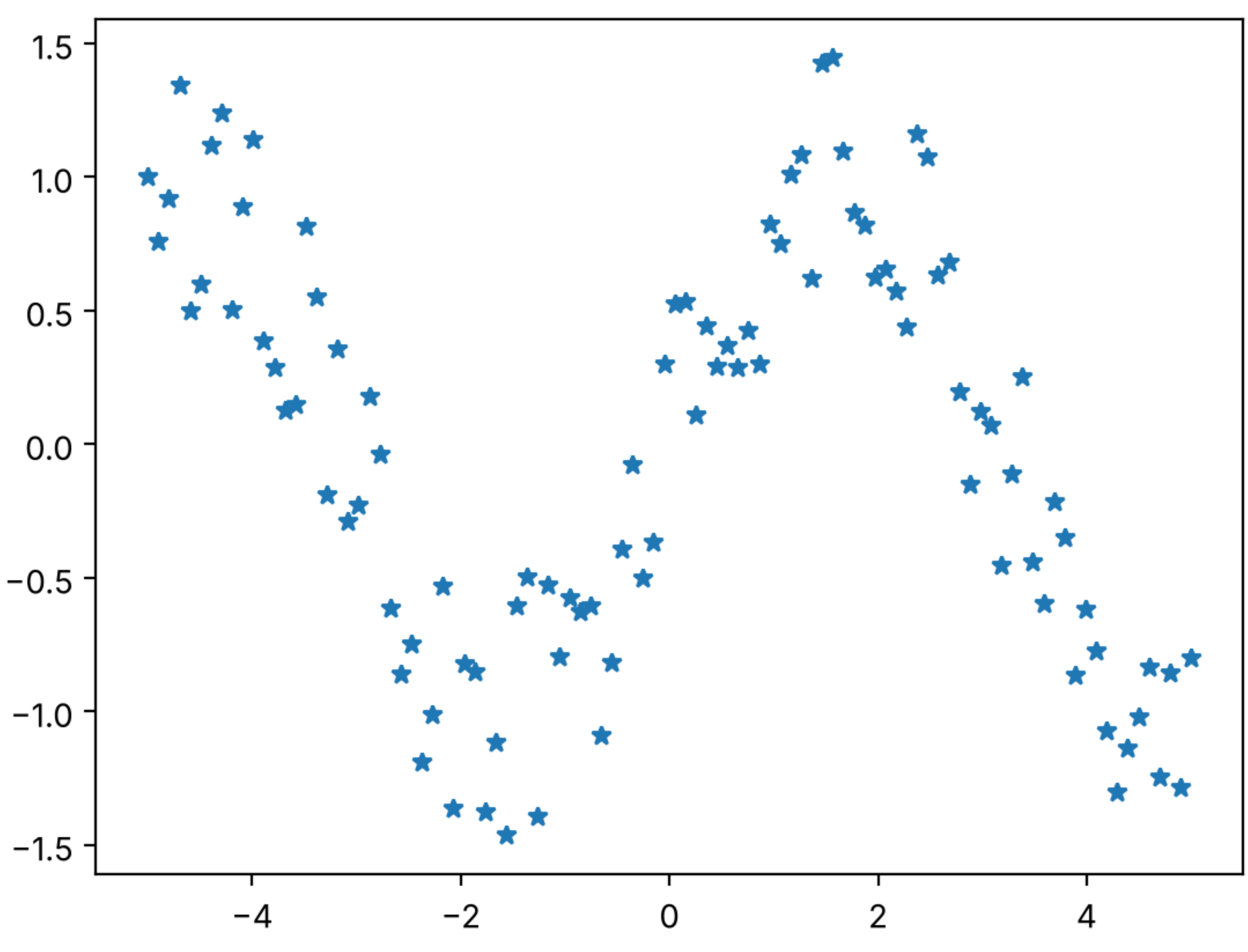
where h is a bandwidth parameter.

Step 1. Simulate a 1-dimensional dataset

[1]:

```python
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
%matplotlib inline

np.random.seed(100)

### Step 1: Simulate a simple 1D data ###
xTrain = np.expand_dims(np.linspace(-5, 5, 100), 1)  # 100*1
yTrain = np.sin(xTrain) + 0.5*np.random.uniform(-1, 1, size=xTrain.shape) ## 100 *1

print('xTrain shape', xTrain.shape, 'yTrain shape', yTrain.shape)
plt.plot(xTrain, yTrain, '*')
plt.show()
```

```
xTrain shape (100, 1) yTrain shape (100, 1)
```



Now we have a dataset with 100 training data points. Let us calculate the kernel function.

Step 2. Kernel function

Your task is to complete the following rbf_kernel function that takes two sets of points X (of size n) and X' (of size m) and the bandwidth h and outputs their pairwise kernel matrix $K = [k(x_i, x_j)]_{ij}$, which is of size $n \times m$. We will represent input data as matrices, with $X = [x_i]_{i=1}^n \in R^{n\times 1}$ denoting the input features and $Y = [y_i]_{i=1}^n \in R^{n\times 1}$ the input labels.)

[2]:

```python
"""
    calculating kernel matrix between X and Xp
"""
from scipy.spatial.distance import cdist

def rbf_kernel(X, Xp, h):
    # X: n*1 matrix
    # Xp: m*1 matrix
    # h: scalar value

    # K[i, j] = exp(-||x_i - x'_j||^2 / (2 h^2))
    sq_dist = cdist(X, Xp, 'sqeuclidean')   # n*m squared Euclidean distances
    K = np.exp(-sq_dist / (2.0 * h**2))

    return K #n*m



### evaluation: if your implementation is correct, you should expect the output is a 2X3 matrix
# [[0.60653066 1.         0.60653066]
# [0.13533528 0.60653066 1.        ]]
k_test = rbf_kernel(np.array([[2],[1]]), np.array([[3],[2],[1]]), 1)
print(k_test)
```

```
[[0.60653066 1.         0.60653066]
 [0.13533528 0.60653066 1.        ]]
```

Step 3. The median trick for bandwidth

The choice of the bandwidth h A common way to set the bandwidth h in practice is the so called median trick, which sets h to be the median of the pairwise distance on the training data, that is

$$h_{med} = median(\{||x_i - x_j|| : i \neq j, \quad i, j = 1, \ldots, n\}).$$

- Task: Complete the median distance function.

[3]:

```python
from scipy.spatial import distance

def median_distance(X):
    # X: n*1 matrix

    dist = distance.cdist(X, X, 'euclidean')   # n*n pairwise distances
    off_diag = [dist[i, j] for i in range(len(X)) for j in range(len(X)) if i != j]
    h = np.median(off_diag)

    return h


### Test your functions
#evaluation: if your implementation is correct, your answer should be [2.0]
h_test = median_distance(np.array([[1],[2],[4]]))
print(h_test)
```

```
2.0
```

Step 4. Kernel regression

The weights w_i are estimated by minimizing a regularized mean square error:

$$\min_w \left(\sum_{i=1}^{n}(y_i - f(x_i; w))^2\right) + \beta w^\top K w,$$

where w is the column vector formed by $w = [w_i]_{i=1}^n$ and K is the kernel matrix.

- Please derive the optimal solution of w using matrix inversion (no need to show the work)
- Complete the following function to implement the calculation of w

[4]:

```python
def kernel_regression_fitting(xTrain, yTrain, h, beta=1):
    # X: input data, numpy array, n*1
    # Y: input labels, numpy array, n*1

    # W = (K + beta*I)^{-1} Y
    K = rbf_kernel(xTrain, xTrain, h)                       # n*n
    W = np.linalg.solve(K + beta * np.eye(len(xTrain)), yTrain)   # n*1

    return W

### evaluating your code, the shape should be (100, 1) (check the values yourself)
h = median_distance(xTrain)
W_test = kernel_regression_fitting(xTrain, yTrain, h)
print(W_test.shape)
```

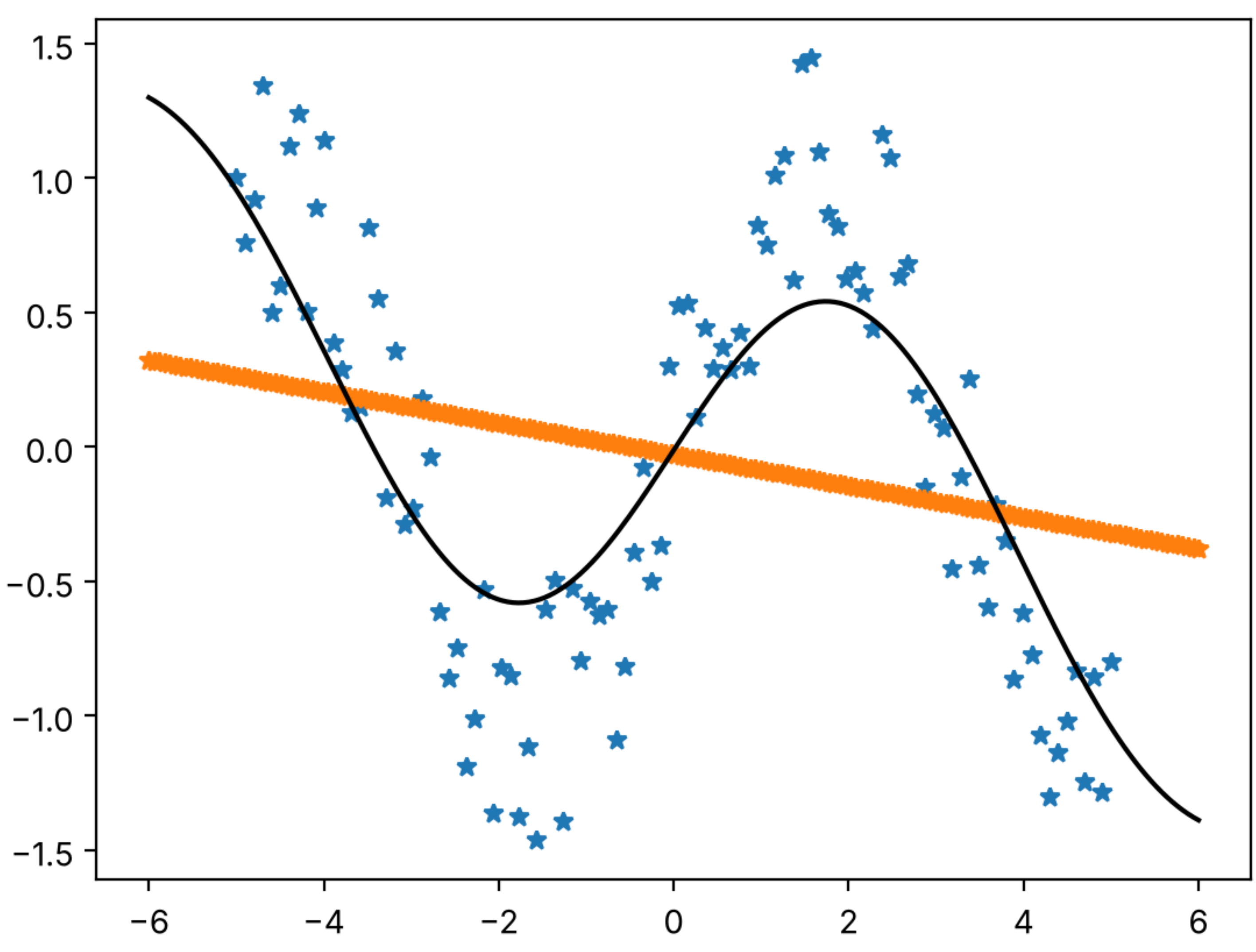
```
(100, 1)
```

Step 5. Evaluation and Cross Validation

We now need to evaluate the algorithm on the testing data and select the hyperparameters (bandwidth and regularization coefficient) using cross validation

[5]:

```python
# Please run and read the following base code

def kernel_regression_fit_and_predict(xTrain, yTrain, xTest, h, beta):

    #fitting on the training data
    W = kernel_regression_fitting(xTrain, yTrain, h, beta)

    # computing the kernel matrix between xTrain and xTest
    K_xTrain_xTest = rbf_kernel(xTrain, xTest, h)

    # predict the label of xTest
    yPred = np.dot( K_xTrain_xTest.T, W)
    return yPred

# generate random testing data
xTest = np.expand_dims(np.linspace(-6, 6, 200), 1) ## 200*1


beta = 1.
# calculating bandwidth
h_med = median_distance(xTrain)
yHatk = kernel_regression_fit_and_predict(xTrain, yTrain, xTest, h_med, beta)


# we also add linear regression for comparison
from sklearn.linear_model import LinearRegression
lr = LinearRegression()
lr.fit(xTrain, yTrain)
yHat = lr.predict(xTest) # prediction

# visualization
plt.plot(xTrain, yTrain, '*')
plt.plot(xTest, yHat, '*')
plt.plot(xTest, yHatk, '-k')
plt.show()
```



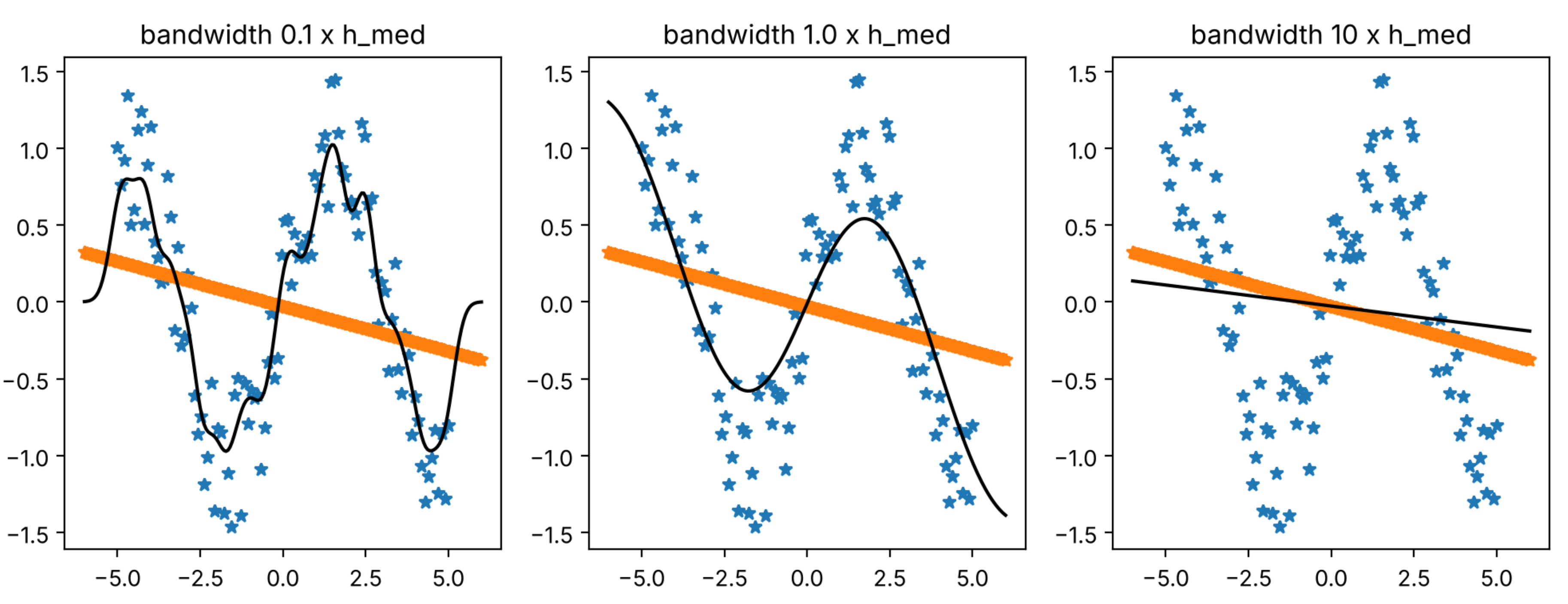
Step 5.1. Impact of bandwidth

Run the kernel regression with regularization coefficient $\beta = 1$ and bandwidth $h \in \{0.1h_{med}, h_{med}, 10h_{med}\}$.

- Task: Show the curve learned by different h. Comment on how h influences the smoothness of the curve.

[6]:

```python
### fitting on the training data ###
beta = 1.

plt.figure(figsize=(12, 4))
for i, coff in enumerate([0.1, 1., 10]):
    plt.subplot(1, 3, i+1)

    yHatk_i = kernel_regression_fit_and_predict(xTrain, yTrain, xTest, coff * h_med, beta)

    # visualization
    plt.plot(xTrain, yTrain, '*')
    plt.plot(xTest, yHat, '*')
    plt.plot(xTest, yHatk_i, '-k')
    plt.title('bandwidth {} x h_med'.format(coff))

plt.show()
```



Smaller bandwidths produce more flexible, less smooth curves that capture local variations, while larger bandwidths produce smoother curves that may underfit the data. This is visible above: with $h = 0.1h_{med}$ the curve follows the sine shape closely and shows small wiggles from the noise, with $h = h_{med}$ it is smooth but the peaks are already flattened, and with $h = 10h_{med}$ it collapses to almost a straight line similar to the linear-regression fit.

Step 5.2. Cross Validation (CV)

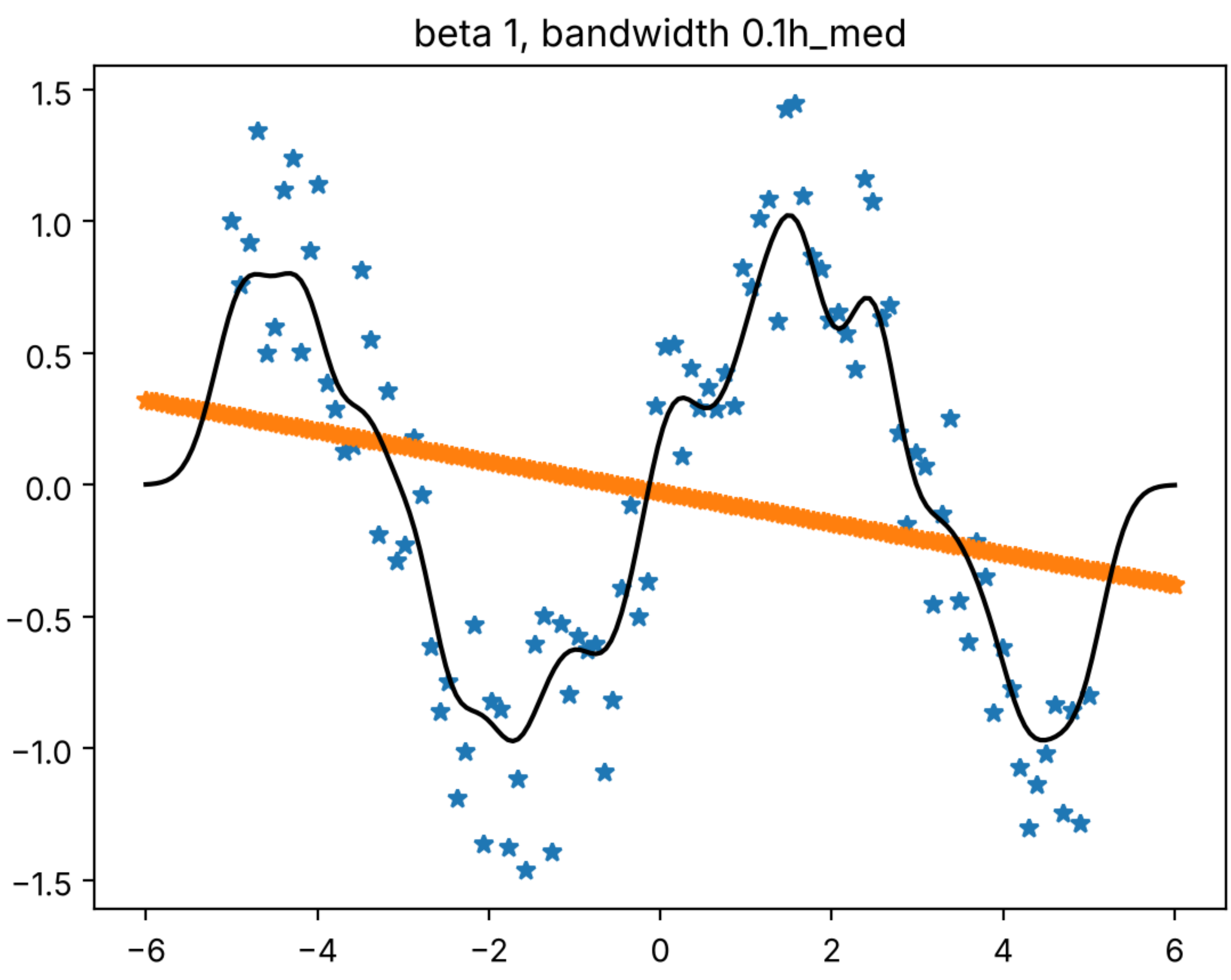
Use 5-fold cross validation to find the optimal combination of h and β within $h \in \{0.1h_{med}, h_{med}, 10h_{med}\}$ and $\beta \in \{0.1, 1\}$.

- Task: complete the code of cross validation and find the best h and β. Plot the curve fit with the optimal hyperparameters.

[7]:

```python
best_beta, best_coff = 1., 1.
best_mse = 1e8
for beta in [0.1, 1]:
    for coff in [0.1, 1., 10.]:
        # 5-fold cross validation
        max_fold = 5
        mse = []
        for i in range(max_fold):

            idx = np.arange(len(xTrain))
            trnIdx = idx[idx % max_fold != i]
            testIdx = idx[idx % max_fold == i]

            i_xTrain, i_yTrain = xTrain[trnIdx], yTrain[trnIdx]
            i_xValid, i_yValid = xTrain[testIdx], yTrain[testIdx]

            h = coff * median_distance(i_xTrain)
            i_yPred = kernel_regression_fit_and_predict(i_xTrain, i_yTrain, i_xValid, h, beta)
            mse.append((i_yValid - i_yPred)**2)

        mse = np.mean(mse)
        print('beta = {:<4} h = {:>4}*h_med   CV MSE = {:.4f}'.format(beta, coff, mse))

        # keep track of the combination with the best MSE
        if mse < best_mse:
            best_beta, best_coff = beta, coff
            best_mse = mse

print('Best beta', best_beta, 'Best bandwidth', '{}*h_med'.format(best_coff), 'mse', best_mse)

# bandwidth
h = best_coff * median_distance(xTrain)
yHatk_i = kernel_regression_fit_and_predict(xTrain, yTrain, xTest, h, best_beta)

# visualization
plt.plot(xTrain, yTrain, '*')
plt.plot(xTest, yHat, '*')
plt.plot(xTest, yHatk_i, '-k')
plt.title('beta {}, bandwidth {}h_med'.format(best_beta, best_coff))
plt.show()
```

```
beta = 0.1  h =  0.1*h_med   CV MSE = 0.1164
beta = 0.1  h =  1.0*h_med   CV MSE = 0.1137
beta = 0.1  h = 10.0*h_med   CV MSE = 0.5922
beta = 1    h =  0.1*h_med   CV MSE = 0.1117
beta = 1    h =  1.0*h_med   CV MSE = 0.1924
beta = 1    h = 10.0*h_med   CV MSE = 0.6003
Best beta 1 Best bandwidth 0.1*h_med mse 0.11166229355896191
```



Using 5-fold cross-validation, the best parameters were $\beta = 1$ and $h = 0.1h_{med}$, with an average MSE of 0.111662.